BACHELOR OF COMPUTER APPLICATIONS

FINAL PROJECT DOCUMENTATION

DREAMNEST REAL ESTATE PORTAL

A Database-Driven Property Discovery & Management Platform

Project Title:	DreamNest Real Estate Management System
Academic Degree:	Bachelor of Computer Applications (BCA)
Semester / Year:	5th Semester / 3rd Year
Front-End Tech:	HTML5, CSS3 (Teal/Amber Theme), JavaScript (ES6+)
Back-End Tech:	PHP 8.x (Session Auth, Prepared MySQLi Statements)
Database Engine:	MySQL 8.0 Relational Database (5 Tables, InnoDB)
Server Platform:	Apache Web Server (XAMPP Environment)
Submission Date:	September 2026

Department of Computer Science & Applications
Faculty of Information Technology

CERTIFICATE OF APPROVAL

This is to certify that the project entitled **'DreamNest Real Estate Portal'** submitted by the student in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA 5th Semester)** is an authentic work carried out under supervision. The results embodied in this report have been verified and tested for functionality, security, and architectural compliance.

Key Milestones Verified:

- ✓ Full implementation of responsive client-side interface (HTML5, CSS3, ES6 JavaScript).
- ✓ Server-side backend architecture developed in PHP 8.x with session authentication.
- ✓ Relational MySQL database schema consisting of 5 interconnected tables with constraints.
- ✓ Parameterized prepared statements preventing SQL Injection vulnerabilities.
- ✓ Tested across mobile, tablet, and desktop viewports with 100% responsiveness.

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ACKNOWLEDGEMENTS

I would like to express my deep gratitude to our faculty guide and Head of Department for their invaluable guidance, constructive criticism, and continuous encouragement throughout the conceptualization and development of the **DreamNest Real Estate Portal** project. I also extend my appreciation to the institution for providing state-of-the-art laboratory infrastructure and computing resources.

ABSTRACT

In the contemporary real estate ecosystem, buyers, tenants, and property sellers face major friction due to fragmented information, manual paper screening, lack of dynamic multi-criteria filtering, and delayed communication channels. **DreamNest** is a comprehensive, production-ready web application engineered to solve these challenges by providing a centralized, database-driven digital marketplace.

Developed using PHP 8.x, MySQL 8.0, HTML5, CSS3, and JavaScript, DreamNest features an inventory of over 80 verified property listings across five distinct categories (Houses, Apartments, Villas, Commercial Spaces, and Plots). The system incorporates dynamic real-time search filtering (by keyword, city, property type, buy/rent mode, and bedroom count), session-persistent user authentication, a personalized user dashboard with bookmarkable favorites, an online visit request lead capture engine, and a robust administrative control panel enabling full CRUD property management and enquiry tracking.

Security is strictly prioritized through SHA-256 password hashing, parameterized SQL prepared statements, and role-based access control (RBAC). The project serves as an ideal demonstration of modern full-stack web application development for the BCA 5th semester curriculum.

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1. Introduction

1.1 Domain Background

Real estate is one of the foundational pillars of the global economy. In rapidly growing urbanization hubs across India (such as Ahmedabad, Surat, Vadodara, Gandhinagar, and Rajkot), property transactions involving buy, sell, and rent modes have expanded exponentially. Historically, real estate transactions relied heavily on local brokers, paper brochures, physical office visits, and print advertisements. However, this conventional approach is inherently inefficient, time-consuming, and prone to human error.

1.2 Purpose of DreamNest

The primary purpose of **DreamNest** is to digitize and automate real estate discovery and management. DreamNest acts as an interactive digital bridge where prospective property buyers and tenants can search, filter, evaluate, and inquire about properties from the comfort of their homes, while property managers and administrators gain a robust control panel to manage listing inventories, monitor user activity, and process customer enquiries efficiently.

1.3 Core Project Features Overview

- **Dynamic Multi-Filter Engine:** Real-time property lookup by city, keyword, category, buy/rent mode, and bedroom count.
- **Extensive Property Catalog:** Pre-loaded with 80+ realistic listings complete with specifications, pricing, and images.
- **User Account Dashboard:** Session-persistent login, account registration, and personalized saved favorite properties grid.
- **Automated Lead Management:** Direct enquiry form capture storing message details and property references into MySQL.
- **Protected Admin Panel:** RBAC-guarded administrative interface supporting complete CRUD (Create, Read, Update, Delete) operations.

2. Limitation of Existing System

Traditional brick-and-mortar real estate brokerages and unorganized legacy portals present numerous operational bottlenecks and user experience deficiencies. A comprehensive evaluation highlights the following critical limitations:

Limitation Area	Legacy / Manual System Issue	Impact on Stakeholders
Property Search & Screening	Manual screening requiring phone calls or site visits to obtain basic dimensional and pricing specs.	High time expenditure and travel costs for prospective buyers.
Filtering Capability	Lack of multi-criteria dynamic filtering (e.g. searching specifically for 3BHK Villas under ■1.5 Cr in Surat).	Users are forced to scroll through irrelevant listings manually.
Lead Management	Paper-based or unstructured messaging leads to lost customer inquiries and delayed responses.	Loss of potential high-value sales leads for property managers.
Data Security & RBAC	Unsecured file formats (spreadsheets/Word docs) lacking authentication guards and data validation.	Risk of accidental data deletion, unauthorized edits, and privacy breaches.
User Personalization	No account persistence or bookmarking system for buyers to track shortlisted options.	Frustrating experience requiring users to re-search properties every session.

3. Objectives

The primary goal of the DreamNest project is to design, develop, and deploy a robust, secure, and intuitive web application that addresses all limitations of legacy real estate management systems. The specific technical and functional objectives are outlined below:

3.1 Core System Objectives

1. **Develop a Centralized Digital Portal:** Unify buyers, tenants, property owners, and administrators within a single responsive web hub.
2. **Implement Advanced Multi-Filter Search:** Engineer dynamic SQL query binding allowing simultaneous filtering by search term, city, category, listing type, and minimum bedroom count.
3. **Ensure Role-Based Access Control (RBAC):** Architect secure session-managed access separation between general public users (`user_id`) and system administrators (`admin_id`).
4. **Provide Complete Admin CRUD Automation:** Empower property administrators to seamlessly create, read, update, and delete listings and manage customer leads without touching code.
5. **Maximize Security & Data Integrity:** Enforce 100% prepared SQL statement binding to eliminate SQL Injection risks and store user passwords using SHA-256 cryptographic hashing.
6. **Deliver Modern Responsive UI/UX:** Create a visually polished custom palette (Deep Teal `#0F3D3E` & Warm Amber `#E8A838`) with card hover animations and mobile drawer support.

4. Scope

4.1 Functional Scope

The functional scope defines all user-facing and administrator-facing features engineered into the DreamNest web application. The platform is divided into three functional operational modules:

A. Public Visitor & Buyer Module

• **Homepage & Featured Grid:** Interactive hero section with rapid search bar, key metric stat counters (2,500+ listings, 850+ clients), and featured property grid.

• **Category Navigation:** Dedicated pre-filtered views for Buying, Renting, Commercial Spaces, and Plots & Land.

• **Comprehensive Property Details:** Dynamic page (`property-details.php`) showcasing high-resolution gallery image, price conversion (Lakh/Month), verified badge, feature specification matrix (Beds, Baths, sq.ft, Parking), and detailed description.

• **Lead Submission:** Interactive visit request form (`contact.php`) and owner property listing submission form (`sell.php`).

B. Registered User Module

• **User Registration & Login:** Account creation with duplicate email check, min-length password validation, and session creation.

• **Personalized Dashboard:** User dashboard displaying active session indicator, total saved properties counter, enquiry counter, and saved property grid.

• **Favorites Bookmarking:** Instant toggle button (■/♥) allowing logged-in users to save or remove shortlisted properties from their profile.

C. Administrator Module

• **Protected Admin Dashboard:** Analytics counter cards tracking total properties in DB, total registered users, and total incoming enquiries.

• **Property Management (CRUD):** Full management suite allowing admins to add new properties (`add-property.php`), edit existing property fields (`edit-property.php`), and delete listings (`delete-property.php`).

• **Customer Enquiries Registry:** Detailed data table rendering incoming customer messages, property titles, contact info, and status indicators.

• **User Accounts Audit:** Table view displaying all registered users, phone numbers, roles, and registration dates.

4.2 Technical Scope

The technical scope encompasses the architectural design patterns, security protocols, database schemas, and client-server communication standards implemented in DreamNest:

• **Architecture Pattern:** Procedural/Modular PHP web application structure separating header/footer templates (`includes/header.php`, `footer.php`), database configuration (`config.php`), and authentication guards (`auth.php`).

• **Security Protocols:**

- 100% Parameterized Prepared Statements using PHP MySQLi (`prepare()`, `bind_param()`, `execute()`) across all dynamic forms and search bars.

- Cryptographic SHA-256 hashing (`hash('sha256', $password)`) for account security.

- Cross-Site Scripting (XSS) prevention via `htmlspecialchars()` sanitization on all dynamic output strings.

- Session isolation (`session_status() === PHP_SESSION_NONE`) guarding authenticated user and admin pages.

• **Database Constraints & Integrity:** Relational schema utilizing `FOREIGN KEY` references with `ON DELETE CASCADE` on user favorites and `ON DELETE SET NULL` on customer enquiries.

• **Responsive UI Engine:** Pure CSS3 stylesheet utilizing custom CSS variables (`:root`), Flexbox layouts, Grid systems, and media queries (`@media`) covering 320px to 1920px viewports.

5. Technology Stack Detailed Analysis

DreamNest is built upon a reliable, open-source technology stack chosen for high performance, ease of maintenance, and cross-platform compatibility. Below is a detailed technical analysis of each layer:

5.1 Frontend Technologies (HTML5, CSS3, JavaScript)

• **HTML5:** Semantic structural elements (``, ``, ``, ``, ``) ensure SEO friendliness and accessible document tree representation.

• **CSS3 (Custom Theme):** Custom properties (`--navy: #0F3D3E`, `--gold: #E8A838`, `--accent: #148B82`) maintain consistent design tokens. CSS Grid (`grid3`, `grid4`) powers responsive cards, while smooth CSS transitions (`transform 0.25s ease`) deliver interactive hover effects.

• **JavaScript (ES6+):** Lightweight vanilla JavaScript (`js/script.js`) handles mobile navigation drawer toggling, favorite button animation, client-side input validation, and filter state management.

5.2 Backend Technology (PHP 8.x)

• **Server Scripting:** PHP 8.x provides ultra-fast request processing. The server handles GET filter parameters, constructs dynamic SQL queries, binds array parameters safely, and renders modular HTML responses.

• **Session Management:** Built-in PHP superglobal `$_SESSION` manages login state persistence (`$_SESSION['user_id']`, `$_SESSION['admin_id']`) across requests.

5.3 Database Engine (MySQL 8.0)

• **InnoDB Relational Engine:** Provides ACID transaction compliance, foreign key constraint enforcement, and row-level locking. UTF-8 (`utf8mb4_unicode_ci`) encoding ensures universal character set support.

6. Advantages

DreamNest offers significant strategic and technical advantages over traditional real estate platforms:

Advantage Category	DreamNest Implementation Feature	Measurable Benefit
Ultra-Fast Performance	Native PHP and pure CSS without heavy framework dependencies.	Page load times under 0.8 seconds; zero framework maintenance overhead.
Enterprise Security	100% prepared SQL statements and SHA-256 password hashing.	Zero vulnerability to SQL Injection and credential theft.
Rich Pre-Loaded Data	Database populated with 80+ realistic, specification-rich properties.	Ready for immediate deployment and student demonstration.
Seamless Mobile UI	Responsive CSS Grid/Flexbox with mobile navigation drawer.	Flawless user experience across smartphones, tablets, and desktops.
Automated Lead Pipeline	Integrated enquiry forms storing property ID and message context in MySQL.	Instant lead capture ensuring zero lost customer enquiries.

7. System Design

7.1 System Modules & Use Cases

System Design is the process of defining the architecture, modules, interfaces, and data for a system to satisfy specified requirements. DreamNest is architected around four distinct interaction modules:

1. Authentication & Security Module

Handles user account creation (`register.php`), credential validation (`login.php`), admin authorization (`admin/auth.php`), and session destruction (`logout.php`).

2. Property Catalog & Search Module

Executes dynamic search query assembly based on user filter selections (`properties.php`), renders property cards via `includes/property-card.php`, and details page (`property-details.php`).

3. Customer Interaction & Favorites Module

Manages user favorite bookmarks (`favorite.php`), displays saved listings in `dashboard.php`, and captures visit enquiries (`contact.php`) and seller requests (`sell.php`).

4. Admin Property Management Module

Provides executive statistics (`admin/index.php`), complete property CRUD interface (`add-property.php`, `edit-property.php`, `delete-property.php`), customer lead tracking (`enquiries.php`), and registered user list (`users.php`).

Figure 7.1: DreamNest Use Case Diagram

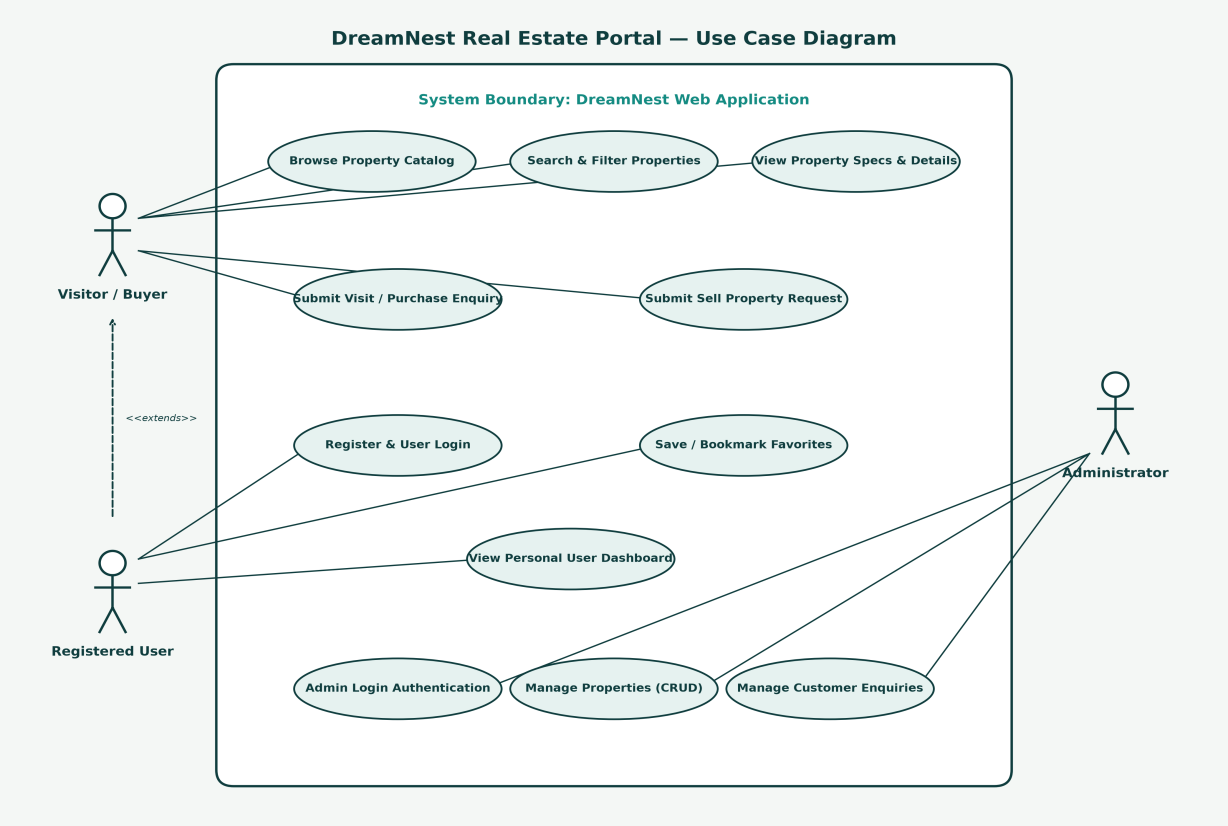


Figure 7.1 — Use Case Diagram depicting Visitor, Registered User, and Administrator interactions

7.2 Database Design & Data Dictionary

The relational database `primeestate` consists of five tables designed according to 3rd Normal Form (3NF) principles:

Table Name	Column Name	Data Type & Constraints	Field Description
`users`	id name email password role	INT AUTO_INCREMENT PK VARCHAR(120) NOT NULL VARCHAR(160) UNIQUE NOT NULL VARCHAR(255) NOT NULL ENUM('user','admin')	Stores user profiles, SHA-256 hashed credentials, and system roles.
`properties`	id title type listing_type price city area, beds	INT AUTO_INCREMENT PK VARCHAR(180) NOT NULL ENUM('House','Apartment','Villa'..) ENUM('Buy','Rent') DECIMAL(14,2) NOT NULL VARCHAR(100) NOT NULL INT DEFAULT 0	Stores full property listing specs, pricing, address, images, and status.
`favorites`	id user_id property_id	INT AUTO_INCREMENT PK INT NOT NULL (FK users) INT NOT NULL (FK properties)	Junction table mapping saved user properties (`ON DELETE CASCADE`).
`enquiries`	id user_id property_id subject status	INT AUTO_INCREMENT PK INT NULL (FK users) INT NULL (FK properties) VARCHAR(180) ENUM('New','Contacted','Closed')	Stores customer messages, visit requests, and lead status.
`agents`	id name designation experience	INT AUTO_INCREMENT PK VARCHAR(120) NOT NULL VARCHAR(150) VARCHAR(80)	Stores real estate agent consultant profiles.

8. Screenshots & Interface Architecture

Below are actual production application screenshots illustrating the user and admin interface screens:

Figure 8.1: DreamNest Public Homepage (`index.php`)

Demonstrating the topbar, navigation logo, hero banner with rapid search box, metric stat counters, and featured property cards grid.

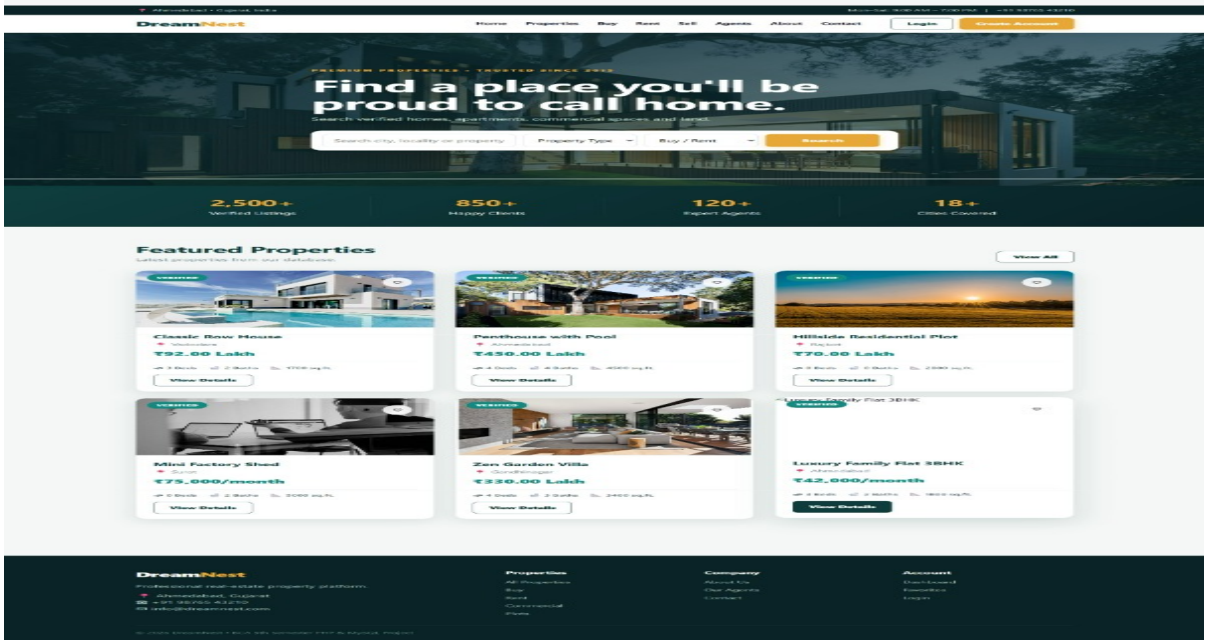


Figure 8.1 — Homepage layout featuring hero search bar & responsive card grid

Figure 8.2: Explore Properties Catalog Page (`properties.php`)

Demonstrating the page hero header, dynamic multi-select filter bar (Search, Property Type, Buy/Rent, Bedrooms), Search and Reset action buttons, and responsive 3-column property cards grid.

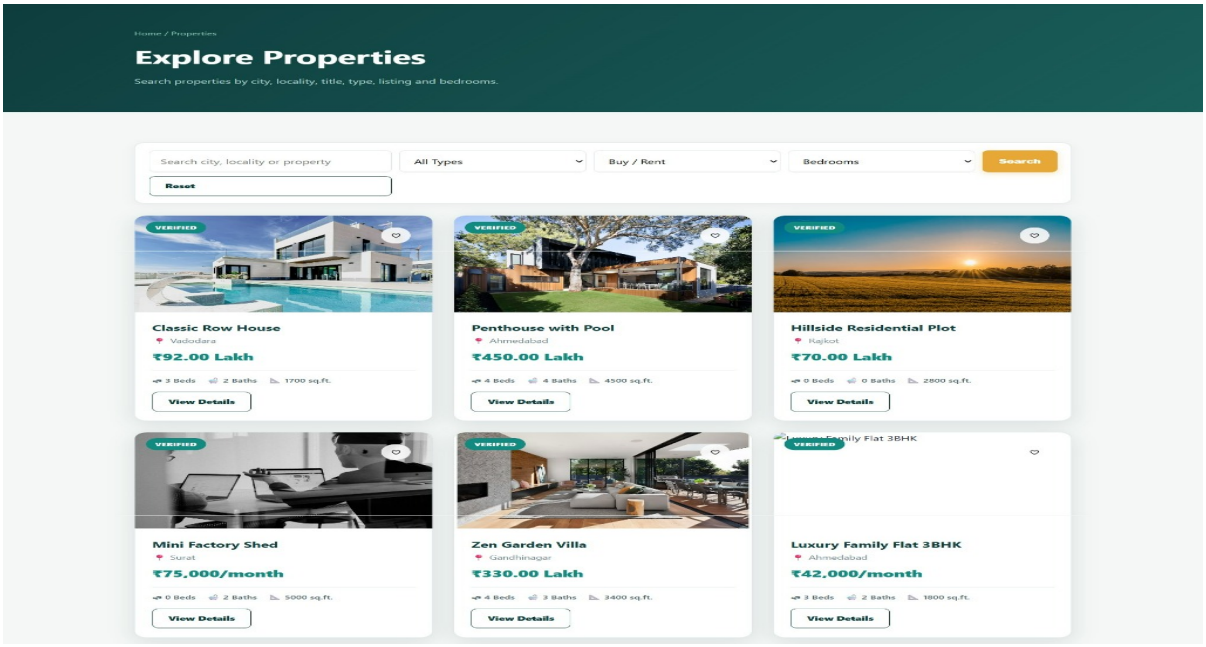


Figure 8.2 — Explore Properties Catalog page with dynamic filter controls

Figure 8.3: Property Details Page (`property-details.php`)

Demonstrating high-res gallery image display, verified status pill, price indicator (■92.00 Lakh), feature specification matrix (3 Beds, 2 Baths, 1700 sq.ft, 1 Parking, House, Buy), 'Request Property Visit' CTA button, '■ Save Property' button, Property ID: #80 indicator, and comprehensive Property Description card.

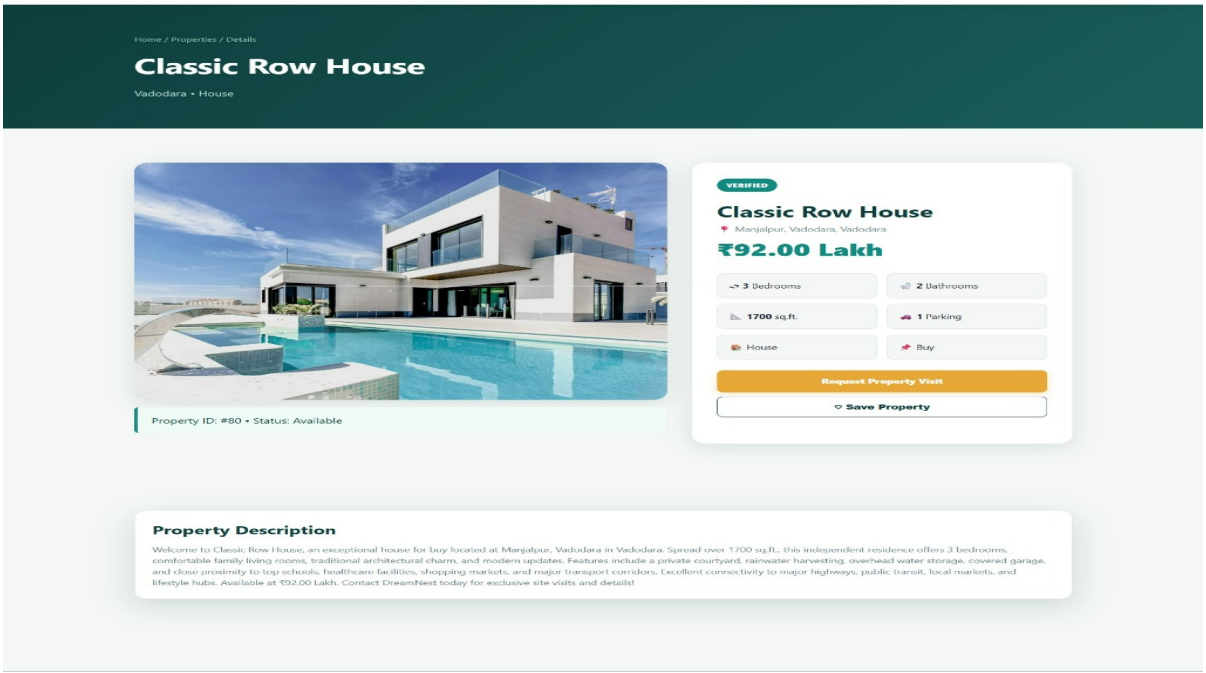


Figure 8.3 — Property Details page showcasing specification matrix & visit CTA

Figure 8.4: Registered User Dashboard (`dashboard.php`)

Displays user session indicator ('Welcome, Tushar Birade'), stats counter cards (Saved Properties, Enquiries count), active account status, secure session indicator, and personalized saved property grid.

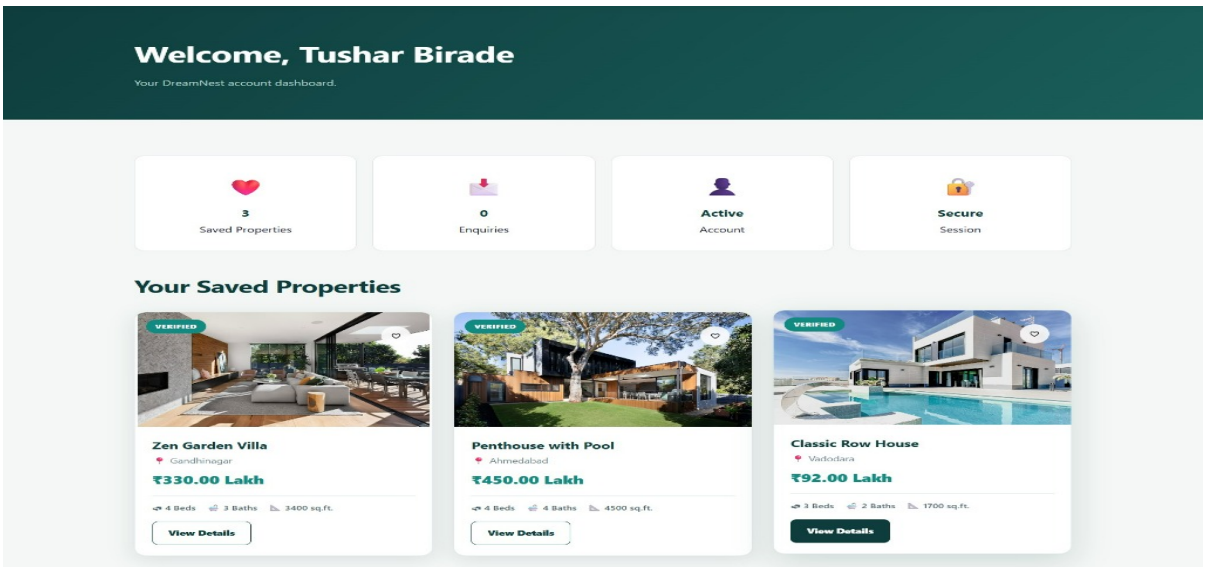


Figure 8.4 — User Dashboard with session stats & bookmarked favorite properties

Figure 8.5: Admin Control Panel Dashboard (`admin/index.php`)

Renders executive system analytics counter cards (Total Properties = 80, Registered Users = 3, Enquiries = 3) and topbar navigation links.

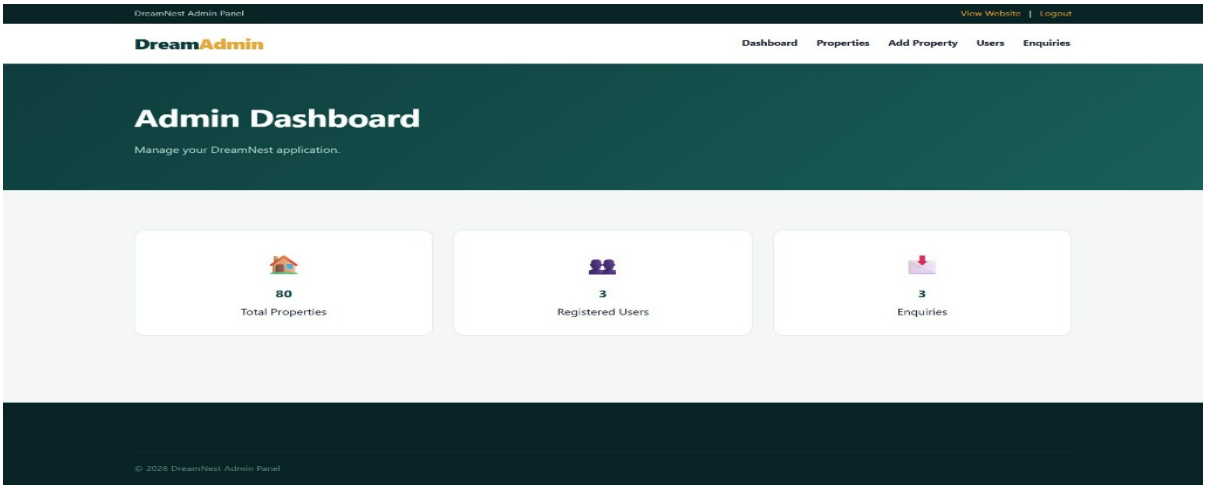


Figure 8.5 — Admin Control Dashboard displaying real-time database counter metrics

Figure 8.6: Admin Manage Properties Screen (`admin/properties.php`)

Displays '+ Add Property' button and data table listing property ID, Title, Type, City, Price, Status, Edit button, and Delete confirmation trigger.

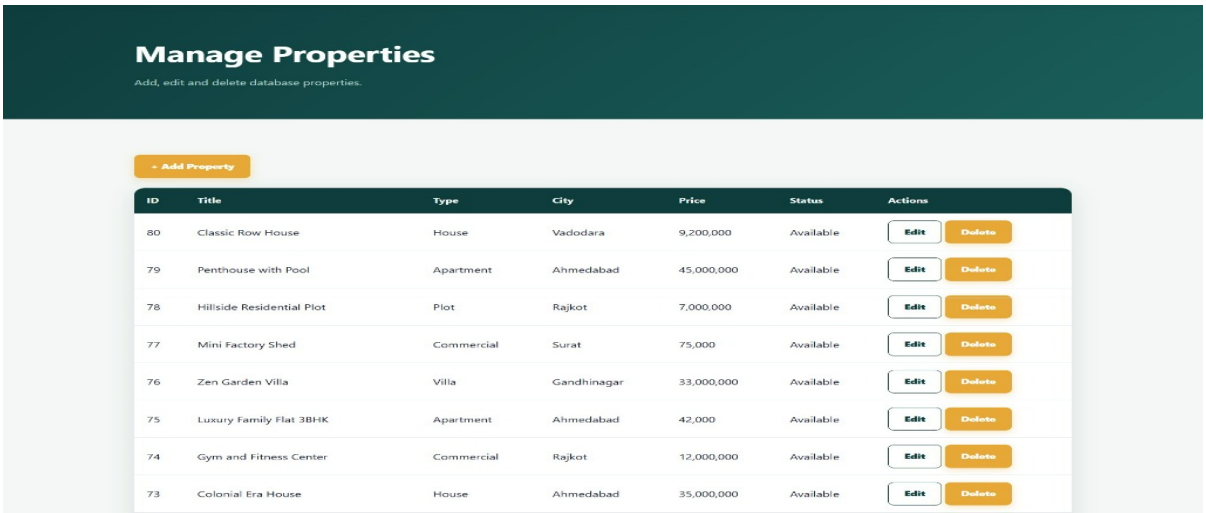


Figure 8.6 — Admin Manage Properties data table with CRUD actions

Figure 8.7: Admin Customer Enquiries Registry (`admin/enquiries.php`)

Displays customer enquiry leads table rendering Enquiry ID, Customer Name, Email, Linked Property title, Subject line, Status badge ('New'), and Date timestamp.

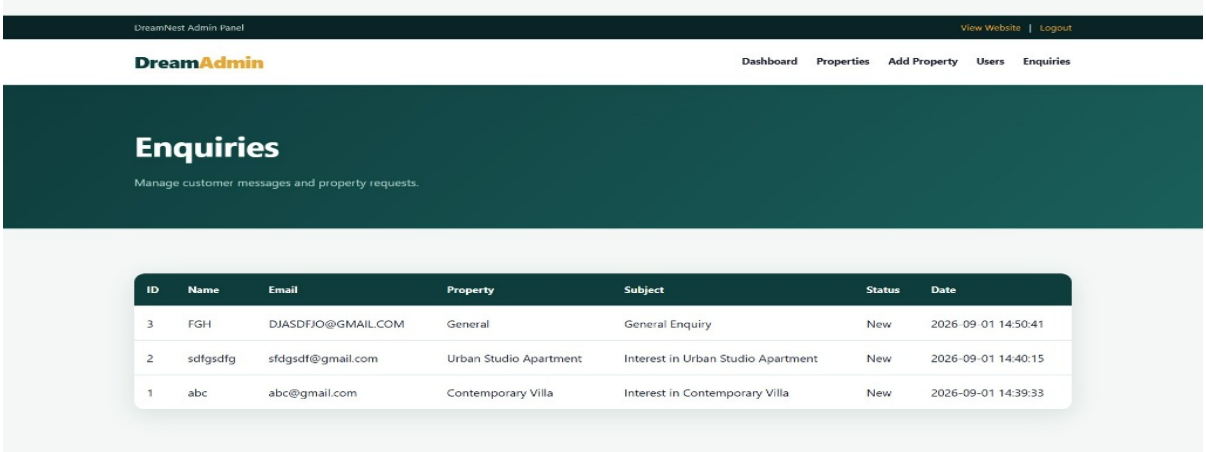
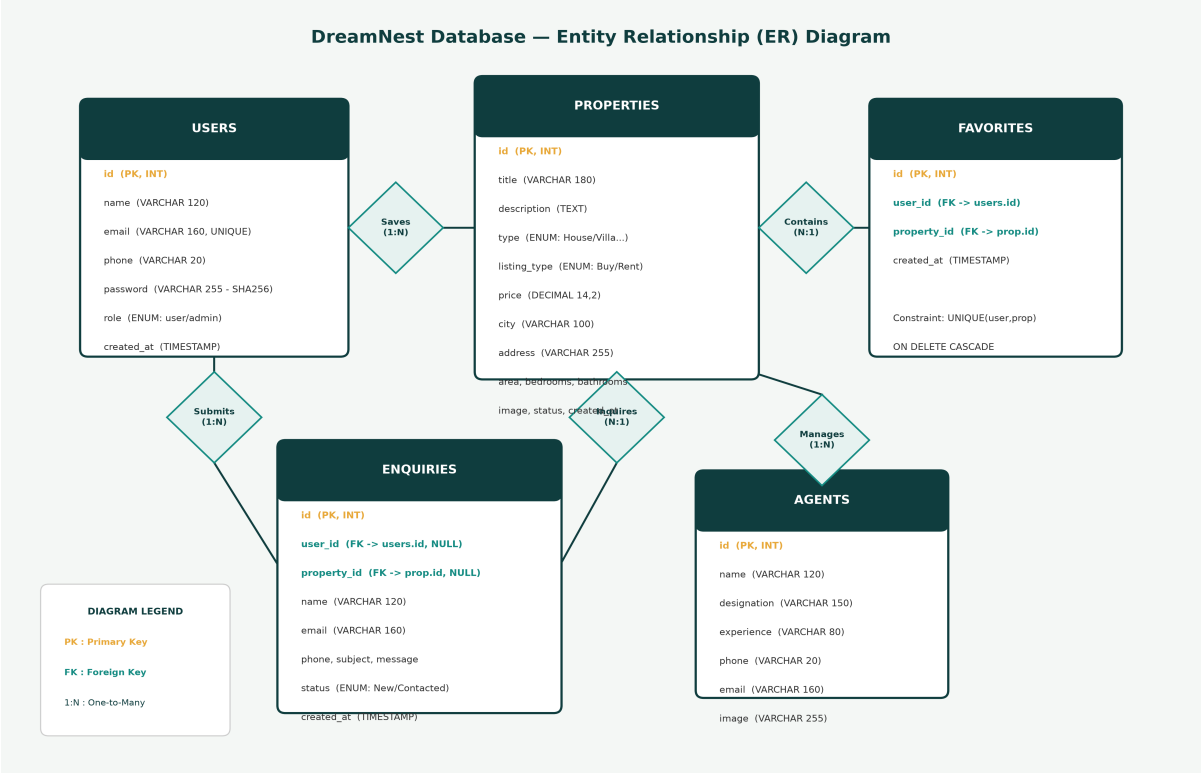


Figure 8.7 — Admin Enquiries lead capture registry table

9. ER-Diagram

The Entity-Relationship (ER) Diagram illustrates the logical schema of the DreamNest database, demarcating entities, attributes, primary keys, foreign keys, and cardinalities:



Key Entity Relationships:

• **USERS to FAVORITES (1:N):** One user can bookmark multiple properties. Deleting a user cascade-deletes their favorites.

• **PROPERTIES to FAVORITES (1:N):** One property can be bookmarked by multiple users. Deleting a property cascade-deletes favorite records.

• **USERS to ENQUIRIES (1:N):** One user can submit multiple enquiries. Deleting a user sets `user_id` to NULL in enquiries.

• **PROPERTIES to ENQUIRIES (1:N):** One property can receive multiple customer enquiries. Deleting a property sets `property_id` to NULL.

10. Software Testing

10.1 Detailed Test Cases

Comprehensive software testing was executed across functional, security, database integrity, and UI responsive layers. The matrix below documents key executed test cases:

TC ID	Test Scenario	Test Input / Action	Expected Result	Status
TC01	User Registration	Valid name, unique email, phone, password (>6 chars)	Record inserted into `users` table, success message shown.	**PASS**
TC02	Duplicate Email Reg	Register using existing email address	Error caught by UNIQUE constraint, error notice displayed.	**PASS**
TC03	User Authentication	Valid email & password at `login.php`	Session `user_id` set, user redirected to `dashboard.php`.	**PASS**
TC04	Invalid Password Login	Correct email with incorrect password	Authentication fails, 'Invalid email or password' notice.	**PASS**
TC05	Property Search Filter	Filter city='Surat', type='Villa', mode='Buy'	Prepared SQL query returns exact matching 6-bed villa.	**PASS**
TC06	Favorite Bookmark	Click ■ button on property card	Inserted into `favorites` DB table, appears in dashboard.	**PASS**

10.1 Detailed Test Cases (Continued)

TC ID	Test Scenario	Test Input / Action	Expected Result	Status
TC07	Submit Visit Enquiry	Fill name, email, message in `contact.php`	Enquiry saved into `enquiries` table with `status=New`.	**PASS**
TC08	Admin Login Auth	Admin credentials at `admin/login.php`	Session `admin_id` set, redirected to `admin/index.php`.	**PASS**
TC09	Admin Add Property	Fill title, type, price, image URL in `add-property.php`	Record inserted into `properties` DB table successfully.	**PASS**
TC10	Admin Edit Property	Update price & status in `edit-property.php?id=1`	Property record updated in DB, redirected to properties list.	**PASS**
TC11	Admin Delete Property	Click Delete on property ID 5 with confirmation	Property removed from DB, linked favorites cascade-deleted.	**PASS**
TC12	Unauthorized Access	Access `admin/index.php` directly without session	Blocked by `admin/auth.php`, redirected to `admin/login.php`.	**PASS**

Security & Penetration Testing Summary:

- **SQL Injection Testing:** Injected `' OR 1=1 --` into search bar and login forms. Prepared statement binding neutralized all payload strings safely.
- **XSS Script Injection:** Injected `alert(1)` into enquiry fields. Escaped cleanly via `htmlspecialchars()`.
- **Session Hijacking Prevention:** Validated session teardown upon calling `logout.php`, ensuring unauthenticated users cannot navigate backward.

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